

ICML 2023: The World Doesn't Stand Still — A Robust Test for Stationarity in Sequential Decision Making

Researchers from the University of Michigan and the London School of Economics propose a doubly robust CUSUM test that decides whether an offline MDP has drifted, and pinpoints where.

Reinforcement learning (RL) is built to find an optimal policy, yet almost every algorithm quietly leans on one assumption: that the world never changes. In statistical terms, the Markov decision process (MDP) has a state-transition function and a reward function that stay fixed over time — the stationarity assumption. It usually sits buried in the derivation, is rarely tested, and yet it largely decides whether the learned policy can be trusted.

Reality is rarely so tidy. In long-horizon settings like mobile health, traffic-signal control, and robotics, the system drifts slowly. Take the Intern Health Study (IHS) that motivates this paper: a micro-randomized trial (MRT) that nudges medical interns in the United States toward healthier habits through phone notifications, whose effect wanes over time. Deploy a policy that treats the world as frozen, and it will send prompts at the wrong moments while long-run reward quietly erodes.

So the question becomes: before trusting an offline-learned policy, can we reliably test whether the stationarity assumption it rests on actually holds? The paper's answer is a model-based, doubly robust statistical test that decides whether an MDP is stationary and locates the change point where it breaks — and stays valid even when the state dimension is high.

Paper: <https://proceedings.mlr.press/v202/wang23ai/wang23ai.pdf>

Code: https://github.com/jtwang95/Double_CUSUM_RL

Why existing tests fall short

Testing stationarity is not new, but prior approaches each have a weak spot. One family needs the true MDP model to be known in advance. Another leans on linear approximations that collapse in high-dimensional state spaces — for example ODCP, proposed by Padakandla et al., and CUSUM-RL, proposed by Li et al. A third learns a separate policy for every time step, wasting samples when the process is actually piecewise homogeneous. The real difficulty is that modern applications tend to be high-dimensional and complex, and drawing trustworthy statistical inference in such settings is hard.

A doubly robust CUSUM test

At the core is a CUSUM (cumulative sum) statistic. For each candidate change point, the test stitches together the transition dynamics before and after it and compares the two: under

stationarity they should agree, while a genuine drift shows up as a difference that peaks near the true change point.

Plugging flexible machine-learning estimators — neural networks, random forests — straight into that comparison introduces heavy plug-in bias. The authors instead add a mean-zero correction term to the naive estimator, yielding a doubly robust estimating function. Doubly robust means the test controls the type-I error as long as at least one of two nuisance models is correctly specified: the transition-function model M_1 or the state-action distribution model M_2 . Sample splitting and cross-fitting then normalize the statistic across candidate change points and test functions, and a Gaussian multiplier bootstrap approximates the null distribution to produce a p-value. The final test statistic is the maximum over time and test functions of the CUSUM-weighted, normalized quantity.

So how robust is that promise that one correct model is enough? Figure 1 offers the most direct evidence. In a controlled synthetic environment, the authors dial the misspecification level of M_1 and M_2 from 0.1 up to 0.9, and repeat every setting 500 times at each error level.

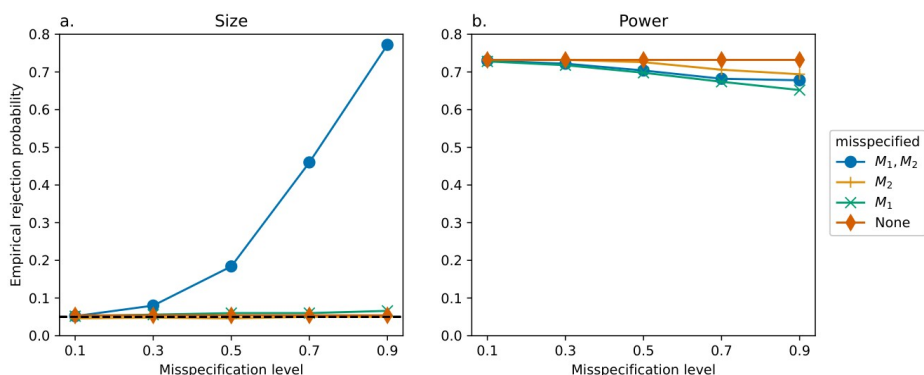


Figure 1. Empirical rejection probability across misspecification levels for four cases (both M_1 and M_2 misspecified, only M_2 , only M_1 , neither). Left is the size under the null; right is the power under the alternative. Results are aggregated over 500 replications.

Read the left panel for size: as long as at least one of M_1 or M_2 is correct, the empirical rejection probability stays pinned near the nominal 5% level ($\alpha = 0.05$); only when both models are misspecified at once does it climb toward 0.9 as the error grows. Read the right panel for power: in every case the test keeps a rejection probability of roughly 0.7 when a change point truly exists. That is exactly what double robustness buys — when one model is wrong, the other covers for it.

The higher the dimension, the bigger the edge

Double robustness handles a possibly-wrong model; what really separates this test from prior work is high dimension. The authors set the state dimension d_S to 1, 10, 20, and 30 and go head-to-head against two baselines, ODCP and CUSUM-RL. The true change point is fixed at $t = 25$, and the test locates it by scanning the interval length κ : when κ is below 25 the null holds and size should be controlled, while once κ crosses 25 the alternative holds and power should rise.

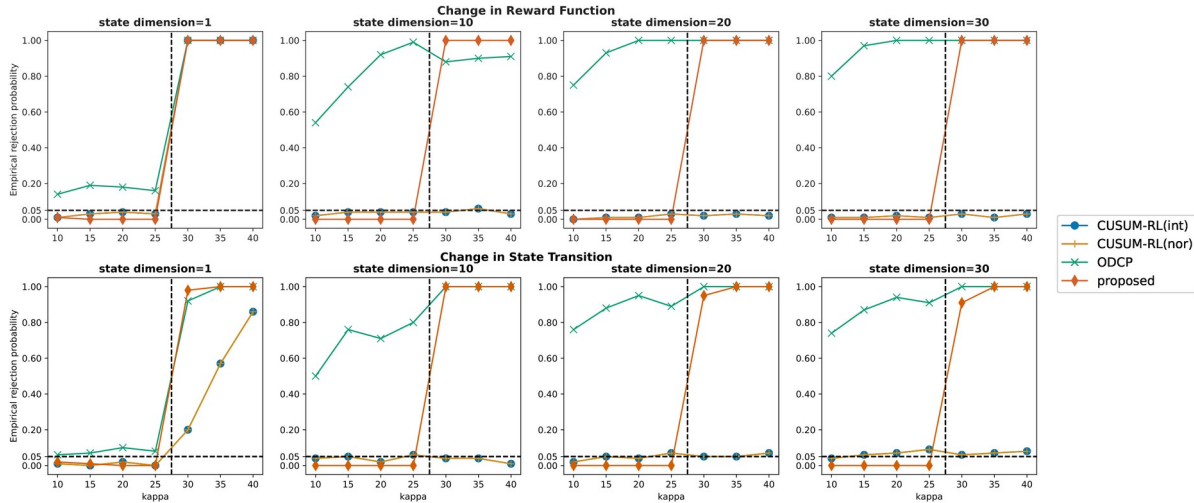


Figure 2. Empirical rejection probability (y-axis) versus the scanning interval length κ under state dimensions $dS = 1, 10, 20, 30$. The top row is a change in the reward function, the bottom row a change in state transition; κ runs from 10 to 40, aggregated over 100 replications.

The result is clean. The proposed test behaves the same way across every dimension from $dS = 1$ to 30: it holds the rejection probability below 0.05 while κ is under 25, then shoots up to 1.0 once κ crosses 25, locking onto the change point. CUSUM-RL, by contrast, is only usable at $dS = 1$ and falls apart as soon as the dimension grows; ODCP inflates the rejection probability far above 0.05 even under the null, so it never controls the type-I error at all. In other words, only this test manages to reject when it should and stay quiet when it should not, even in high dimensions.

Table 1. Qualitative comparison of the three tests in high dimensions.

Method	Type-I control	Highest state dim recovered
Proposed	Yes	30
CUSUM-RL	Yes	1
ODCP	No	-

After the change point is found: how much reward comes back

The value of a test ultimately shows up in the policy. If the change point is located correctly, one can relearn the policy using only the stationary stretch of data after it. In a 4×4 grid world, the authors compare several update strategies: oracle (the true change point is known), random (a random split), slide (a sliding window), and stationary (assume stationarity and use all the data).

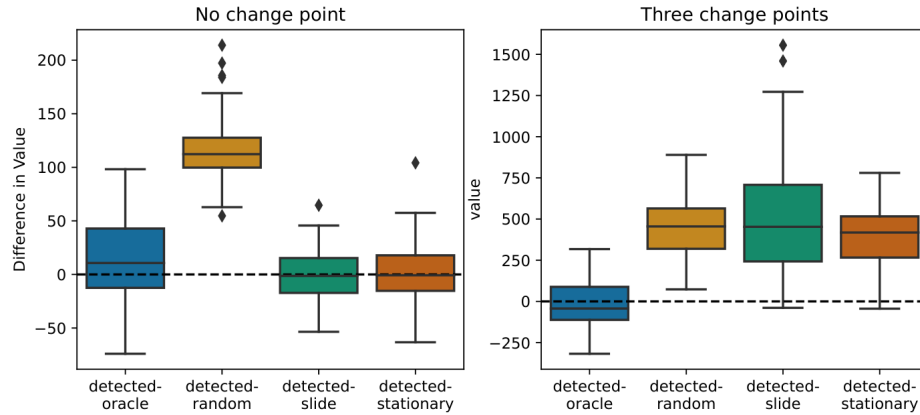


Figure 3. Boxplots of the difference in policy value between the proposed method and four update strategies (oracle, random, slide, stationary). Left is the no-change-point case; right is the case with several change points.

When the data really does contain change points (right panel), the policy learned by the proposed method essentially ties with oracle and clearly beats the random split, the sliding window, and the assume-stationary strategy — the last of which blends heterogeneous pre- and post-change data and ends up with a lower-value policy. And when the data is stationary to begin with, with no change point (left panel), the method does no harm: it lines up with oracle, the sliding window, and the stationary assumption. Detecting and adapting to change points recovers near-oracle reward when it matters, and costs nothing when it does not.

Real data: the Intern Health Study

Finally, the authors apply the test to the real IHS data. The study uses a phone app to send medical interns in the United States reminders that encourage healthy behavior around physical activity, sleep, and mood; the analysis here spans 21 weeks. The interns are split by specialty into Internal Medicine and Family Practice, and κ is scanned from 3 to 18 to compute a p-value for each group.

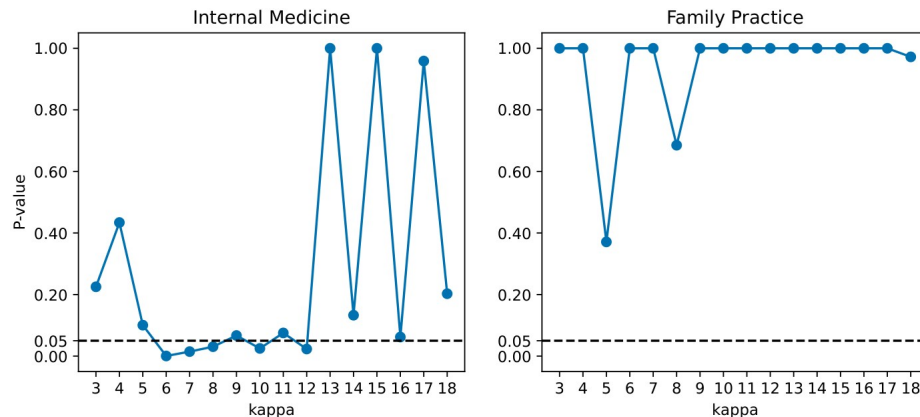


Figure 4. P-values from applying the proposed test to the two specialty groups of the IHS data, as a function of the scanning interval length κ , with κ from 3 to 18. The dashed line marks the 0.05 significance level.

For the Internal Medicine interns (left), the p-value drops below the 0.05 line repeatedly around $\kappa = 16$, indicating that their MDP changed near week 16 — consistent with the clinical

intuition that the nudges' effect wanes over time. For the Family Practice group (right), the p-value stays far above 0.05 throughout, and the test correctly reports no change point. A test that raises the alarm when it should and stays silent when it should not is the hallmark of trustworthy inference.

Takeaways and boundaries

This work marries the flexibility of modern machine learning with the rigor of semiparametric statistics: a doubly robust CUSUM test that reliably signals when an offline RL environment stops being stationary, and holds up even at a state dimension as high as 30. As long as at least one of the two nuisance models is correct, it controls the type-I error at the nominal 5% level ($\alpha = 0.05$). On the theory side, the authors prove size control and double robustness under a bidirectional asymptotic framework, where either the number of trajectories N or the horizon length T may diverge. On the empirical side, the method holds across four numerical studies and the real IHS data, with a typical setting of $N = 100$, $T = 50$, and 5,000 bootstrap samples to approximate the null distribution ($B = 1,000$ for the real data).

The approach does have boundaries: test-based policy relearning is only feasible when the data carries some homogeneity — the change points across trajectories need to overlap, and the MDP must be piecewise constant. But within that regime, test first, then locate, then adapt, and you can reclaim the reward that an assume-stationary or sliding-window strategy simply leaves on the table. For anyone deploying an offline-learned policy into a real system that drifts, this is worth doing first: before trusting a policy, test whether the ground beneath it has really stayed still.

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